Chapter 3
METHODOLOGY

This chapter presents the research methods that will be used in the study. It describes the developmental research design that will guide the development and evaluation of the proposed system. It discusses the system architecture, along with the modeling tools that will be used to represent system functionality, including the data flow diagrams, entity-relationship diagram, and use case diagram. It identifies the sources of data and defines the population, specifying the research locale where the study is conducted. It also outlines the procedures that will be followed during system development and testing, explains the instruments and data collection procedures that will be used to gather information from participants, and presents the statistical tools that will be used for data analysis to ensure accurate and objective interpretation of the findings.

Research Design

This study will employ a mixed-method research design, combining both qualitative and quantitative approaches to support the development and evaluation of the proposed dental health record management system with predictive analytics. The mixed-method approach is selected to ensure that the system design is based on actual operational needs in school dental clinics while also allowing an objective assessment of system quality and predictive performance.

The qualitative component will focus on understanding the current practices, workflows, and challenges in managing dental health records in school settings. Data will be gathered through interviews with school dental personnel, observation of existing record management processes, and review of related documents such as patient treatment records. Particular attention will be given to how patient information is collected, stored, and updated, as well as how dental services such as examination, treatment, and follow-up are managed. These qualitative findings will guide the identification of system requirements, functional modules, and constraints, including issues related to manual record keeping, data accessibility, and monitoring of patient conditions. In addition, the structure and attributes of the dental dataset will be analyzed to determine relevant variables for predicting oral health conditions.

The quantitative component will be applied during the system testing and evaluation phase. Structured questionnaires will be used to collect numerical ratings from users and evaluators based on the ISO 25010:2023 quality model.

By integrating qualitative insights with quantitative evaluation, the mixed-method research design will ensure that the proposed system is both relevant to actual dental clinic operations and objectively evaluated against established software quality standards.

Software Development Methodology

The study will use the Agile Software Development Methodology as the framework for developing Floral, a web-based Dental Health Record Management System with integrated predictive analytics for the school dental clinics of Barangay Tanyag. Agile is an iterative and incremental approach that allows continuous improvement through repeated development cycles and ongoing stakeholder feedback [55].

The development process will be carried out through iterative cycles (sprints), where each phase will be revisited to refine and enhance the system based on evaluation results and feedback from the primary client, the school dentist of Barangay Tanyag, as well as dental aides, clinic staff, and school administrators.

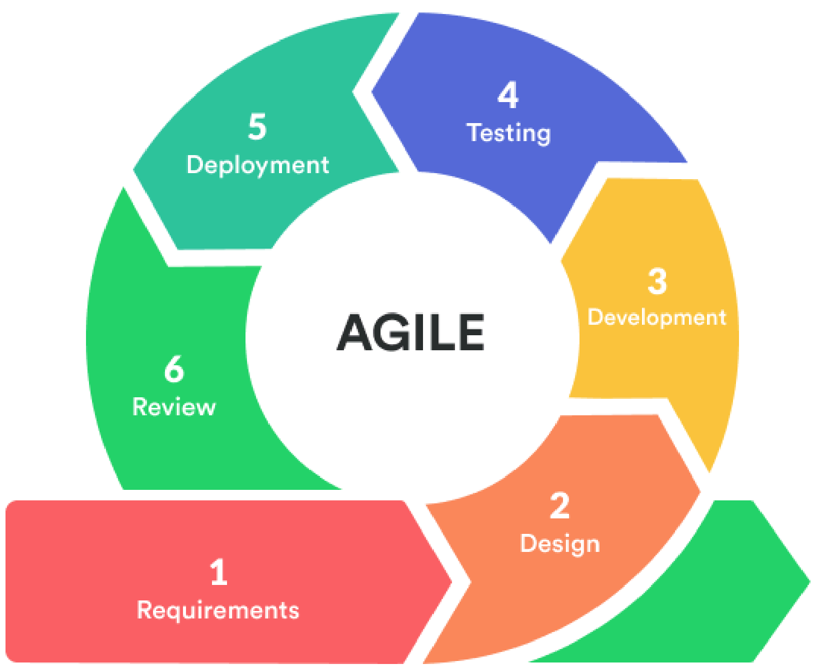

Figure 2. *Agile Methodology, Software Development Life Cycle*

Figure 2 illustrates the iterative nature of the Agile Software Development Life Cycle, where each phase is interconnected and continuously repeated. The process allows the researchers to design, develop, test, and refine the system in incremental cycles. Feedback gathered from each phase is used to improve the system in succeeding iterations, ensuring that the proposed dental health record management system with a predictive component meets user requirements.

Phase 1: Requirements

This phase focuses on identifying the functional and non-functional requirements specific to Floral. Data will be gathered through semi-structured interviews with the school dentist and dental aide of Barangay Tanyag, direct observation of existing record management workflows across Bagong Tanyag Integrated School, Bagong Tanyag Elementary School Annex A, and South Daang Hari Elementary School Main, and review of official DOH clinical documents currently in use, including the Individual Patient Treatment Record (IPTR), the Target Client List for Oral Health Care and Services, the Oral Health Program Reporting Form, and the Section-Level Oral Health Status Tally Form [4].

The structure and attributes of the DOH IPTR dataset will be systematically analyzed to determine the data inputs required for both the record management and predictive analytics components of Floral. Variables to be considered include, but not limited to, patient demographics, medical history, dietary habits, social history, oral health conditions, dental chart entries per tooth, treatment records, and routine preventive care data [1], [2].

The findings that will be collected will define the core functional requirements of the proposed system such as, secure role-based authentication for dentists, dental aides, and school administrators; centralized student dental record management across three schools; digital dental charting; appointment scheduling and monitoring; two-visit routine preventive care (RPC) monitoring; predictive oral health risk classification into High, Medium, or Low risk levels; and automated generation of standardized monthly reports and interactive dashboards aligned with DOH reporting formats [3], [4]. These requirements will be refined in subsequent sprints based on user feedback and system testing outcomes.

Phase 2: Design

In this phase, the overall architecture and module structure of Floral will be designed based on the confirmed requirements. Role-based access control will be designed to restrict system features and data visibility according to user role: Dentist, Dental Aide, School Administrator, and Barangay Health Office staff [26], [27].

System models will be produced to represent the proposed system’s structure and data flows, including the Context Diagram, Top Level Data Flow Diagram, Entity-Relationship Diagram (ERD), and System Architecture [14].

The predictive analytics component will be designed as a structured machine learning pipeline consisting of four stages: (1) data preprocessing, which involves cleaning and normalizing IPTR records; (2) feature engineering, which transforms IPTR attributes such as DMF/dmf index scores, oral health conditions, dietary habits, and medical history into model-ready inputs; (3) model training and comparison of multiple algorithms; and (4) risk classification output generation [7], [8], [9]. The interface for the decision-support module will be designed to present risk classification results alongside condition-appropriate treatment recommendations for dentist review and validation.

Phase 3: Development

Development will proceed sprint by sprint, with each sprint targeting a specific system module: (1) user authentication and role-based access control; (2) student registration and centralized dental record management; (3) digital dental charting (4) appointment scheduling and monitoring with follow-up flagging; (5) two-visit routine preventive care (RPC) monitoring, including oral screening, oral prophylaxis, and fluoride varnish application records [4]; (6) predictive analytics integration; and (7) dashboard and automated report generation aligned with DOH reporting standards [3].

For the predictive analytics module, development will use Python-based machine learning libraries. The pipeline will include data cleaning of historical IPTR records, exploratory data analysis to identify relevant predictive variables, and feature engineering to convert raw dental record attributes into structured model inputs. Multiple classification algorithms will be implemented and compared based on their performance. The algorithm demonstrating the strongest results across accuracy, precision, recall, and F1 score will be selected and integrated into Floral as the active risk classification. Outputs will be passed to the system's decision-support interface, where the dentist reviews and validates treatment recommendations before any clinical action is taken [51], [52].

Phase 4: Testing

This phase will involve both functional and predictive testing of Floral. Alpha Testing and Beta testing will be conducted to verify that individual modules, including dental charting, appointment scheduling, RPC monitoring, and report generation, function correctly both in isolation and in combination [25].

The predictive analytics module will be evaluated using standard classification metrics: accuracy, precision, recall, and F1 score [55]. Performance will be compared across all algorithms using the preprocessed historical IPTR dataset, and the best-performing model will be confirmed for integration. System quality will be assessed using the ISO/IEC 25010:2023 standard [56]. Identified bugs and discrepancies across all modules will be documented and resolved before the next sprint begins, ensuring that the proposed system maintains functional stability and predictive reliability throughout the development cycle.

Phase 5: Deployment

In this phase, the proposed system will be prepared for pilot deployment across the three covered school dental clinics: Bagong Tanyag Integrated School, Bagong Tanyag Elementary School Annex A, and South Daang Hari Elementary School Main. [7], [9].

Role-based user accounts will be configured for the school dentist, dental aides, school administrators, and clinic staff. System access will be set up across devices used at the clinic level [25]. Audit trail functionality will be activated to log all user actions - including record additions, edits, and deletions, across all three school sites, in compliance with data security and privacy requirements [26], [27]. Initial deployment may be followed by further adjustments based on early user feedback prior to formal evaluation.

Phase 6: Review

The final phase will focus on the formal evaluation of the proposed system's software quality and the collection of client's feedback to guide final refinements. The system will be evaluated using a survey questionnaire based on the ISO/IEC 25010:2023 Software Product Quality standard [56].

Quantitative responses collected via a five-point Likert scale [59] will be analyzed using weighted mean, frequency count, and percentage [58]. The total population sampling approach will be applied, as all qualified respondents within each group are accessible and their full participation is feasible [57]. Qualitative feedback gathered through post-evaluation consultations will be used to identify specific areas for system improvement. Findings from this phase will inform the final iteration of Floral's development, including any interface enhancements, performance optimizations, or additional features identified as critical by end users. Recommendations for future development, such as integration with national DOH health databases, expansion to other barangay-level clinics, or enhancement of the predictive model will also be documented in this phase.

System Architecture

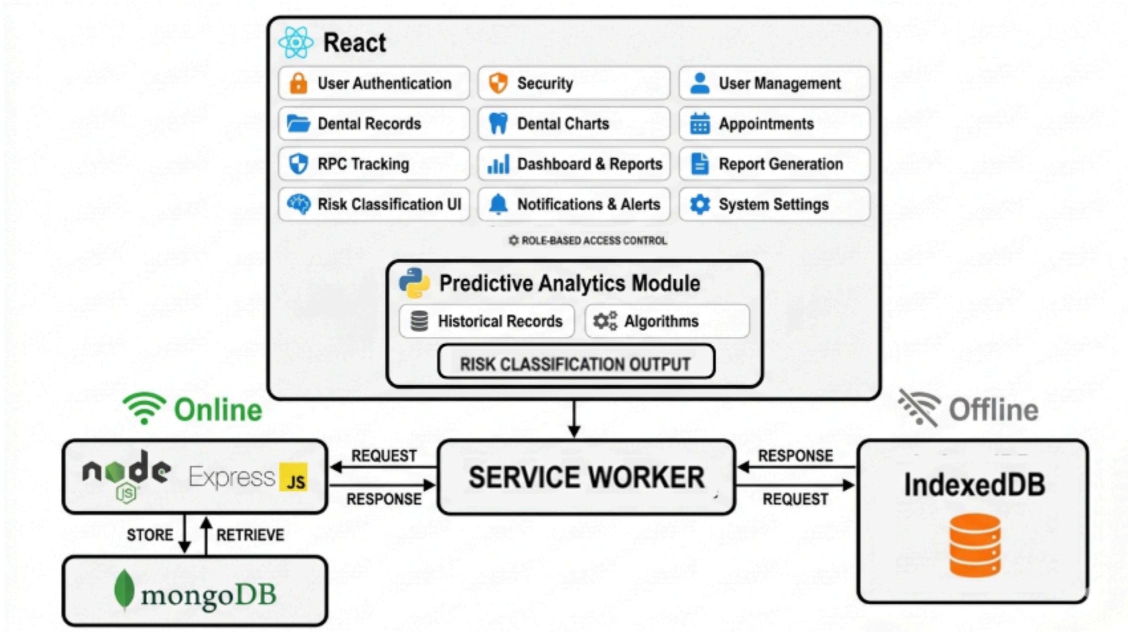

Figure 3. *System Architecture Diagram*

Figure 3 presents the system architecture of the proposed dental health record management system. The architecture illustrates the interaction between the front-end application, backend services, and both online and offline data storage components.

The system utilizes a React-based front-end interface, which provides core functionalities including login, authentication and security, user management, dental records, dental charting, appointment management, RPC tracking, dashboard and reporting, report generation, risk classification interface, and system settings. Access to these features is controlled through role-based access control, supporting different user roles such as Dentist, Dental Aide, School Administrator, and Barangay Health Office staff.

For online operations, the system communicates with a Node.js Express backend, which handles request and response processing. The backend interacts with a MongoDB database for storing and retrieving patient records, treatment data, and system-generated outputs.

To support offline functionality, the system incorporates a Service Worker, which manages offline caching, background synchronization, and request handling. When the system is offline, data is temporarily stored in IndexedDB, which holds offline patient records and a pending synchronization queue. Once connectivity is restored, the Service Worker synchronizes the locally stored data with the backend server and MongoDB database.

This architecture ensures continuous system availability, allowing users to access and update records even without internet connectivity while maintaining data consistency through automatic synchronization when the system returns online.

Context Diagram

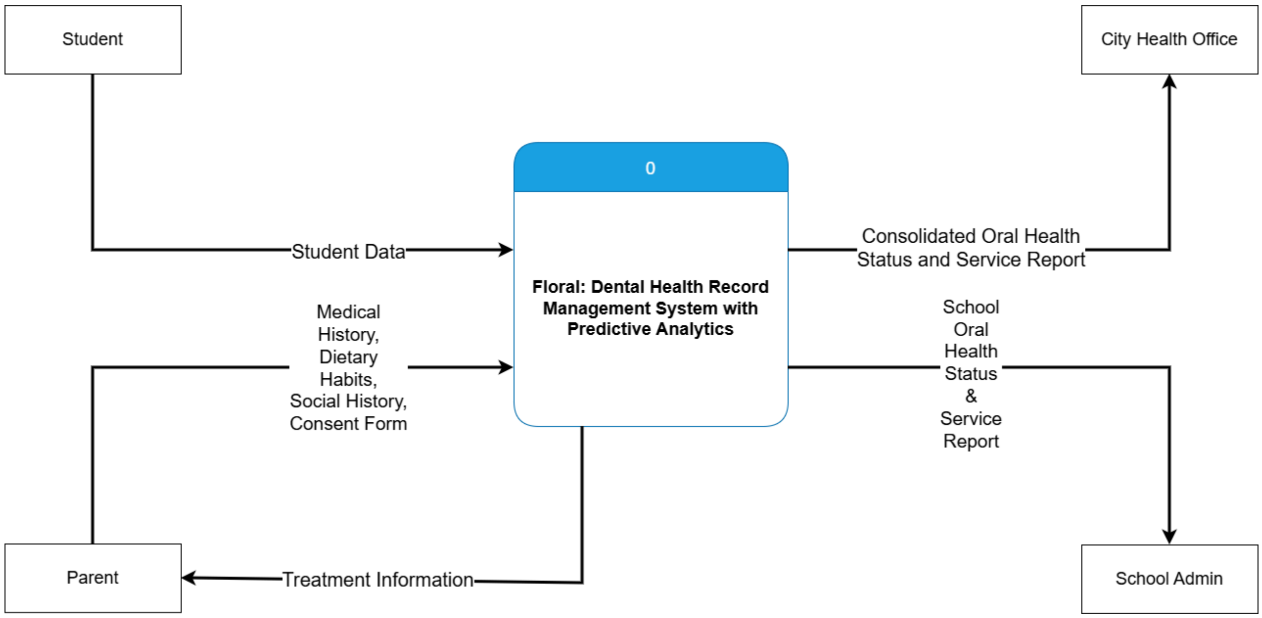

Figure 4. *Context Diagram*

Figure 4 presents the Context Level Diagram of the proposed system. The diagram illustrates the overall boundary of the system and its interactions with four external entities.

The first external entity is the Student, who provides student data and dental data to the system. These inputs include the student's personal information and dental health records needed for processing. In return, the system generates RPC information which is sent back to the student, informing them of their routine preventive care visit schedule.

The second external entity is the Parent, who provides consent form data, medical history data, and dietary and social habits data to the system. These inputs are essential for completing the student's Individual Patient Treatment Record. In return, the system sends treatment information back to the parent, informing them of the dental treatments administered to their child.

The third external entity is the Class Adviser, who provides the student list to the system. This input allows the system to reference and manage the roster of enrolled students across the covered schools.

The fourth external entity is the City Health Office, which receives the Consolidated Oral Health Status and Service Report from the system. This report provides aggregated data on the oral health status and dental services rendered across the three covered school dental clinics, supporting monitoring and decision-making at the city health level.

Top Level Diagram

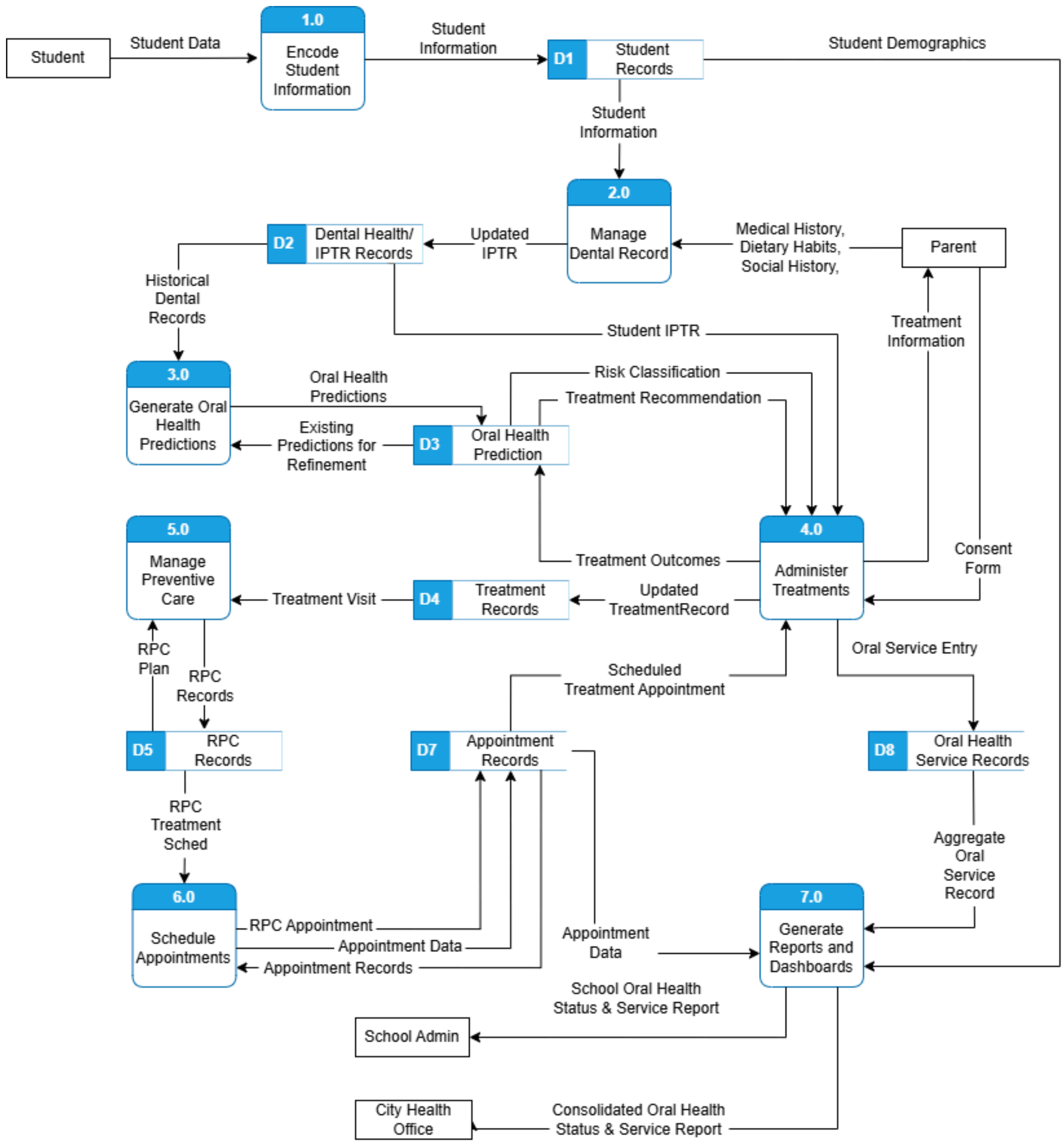

Figure 5. *Top Level Diagram*

Figure 5 illustrates the Top Level (1) Diagram of the proposed system. It shows how information flows between external entities, system processes, and data stores to support student records, dental chart management, predictive analytics, treatment records, preventive care, scheduling, and reporting.

The system will be initiated through **Process 1.0** (Encode Student Information), which serves as the primary entry point for student data into FLORAL. Prior to system implementation, existing paper-based dental records maintained by the school dental clinics are represented as D0 (Existing Paper-Based IPTR Records), which will be encoded into the system during the initial data migration phase. Alongside this, the Class Adviser will provide the student master list, and the Student will provide their personal student data, both of which will be processed and stored in D1 (Student Records). These records will form the core dataset used across the system and will support subsequent processes such as dental charting and reporting.

Process 2.0 (Manage Dental Chart / IPTR) will be performed using student information retrieved from D1 (Student Records). Clinical inputs will be gathered directly from the Student, including dental data collected during the clinic visit, as well as from the Parent, who will provide the consent form, medical history, and dietary and social habits of the student. This process will generate and update the student's Individual Patient Treatment Record (IPTR), which will be stored in and retrieved from D2 (IPTR Records). The updated IPTR will ensure that clinical data remains accurate and continuously maintained across all visits.

Process 3.0 (Generate Oral Health Predictions) will analyze historical records and DMF indices retrieved from D2 (IPTR Records) to produce oral health predictions. These predictions will be stored in and retrieved from D3 (Oral Health Prediction) and will include risk classification and predictive insights about a student's oral condition. The generated predictions will serve as the basis for treatment recommendations and risk assessment in the succeeding process.

Process 4.0 (Administer Treatments) will operationalize clinical interventions guided by risk classifications and treatment recommendations retrieved from D3 (Oral Health Prediction). Treatment details and outcomes will be recorded in D4 (Treatment Records) and will also contribute to D7 (Oral Health Service Records) through oral service entries. Where treatments require parental supervision, a corresponding flag will be passed to Process 6.0 (Schedule Appointments) to ensure proper coordination. Treatment information will also be communicated to the Parent to keep them informed of their child's dental care.

Preventive care will be managed in **Process 5.0** (Manage Preventive Care), which will utilize treatment visit data from D4 (Treatment Records) to develop Routine Preventive Care (RPC) plans. These plans will be stored in and retrieved from D5 (RPC Records). This process will emphasize proactive care by ensuring that students receive continuous monitoring and preventive interventions beyond immediate treatment. RPC information will also be communicated directly to the Student to support awareness and compliance with the two-visit preventive care process.

Scheduling functions will be handled by **Process 6.0** (Schedule Appointments), which will integrate RPC next visit schedules from D5 (RPC Records) and maintain appointment data in D6 (Appointment Records). This process will ensure proper coordination of treatment visits, including those flagged as requiring parental supervision from Process 4.0.

Process 7.0 (Generate Reports and Dashboards) will consolidate data from multiple sources, including appointment records from D6 (Appointment Records) and aggregate service data from D7 (Oral Health Service Records). It will produce consolidated oral health status and service reports, which will be submitted to the City Health Office to support monitoring, resource allocation, and evidence-based decision-making at the city level.

Child Diagram?? (Figure 5.1 or 6)

Entity-Relationship Diagram

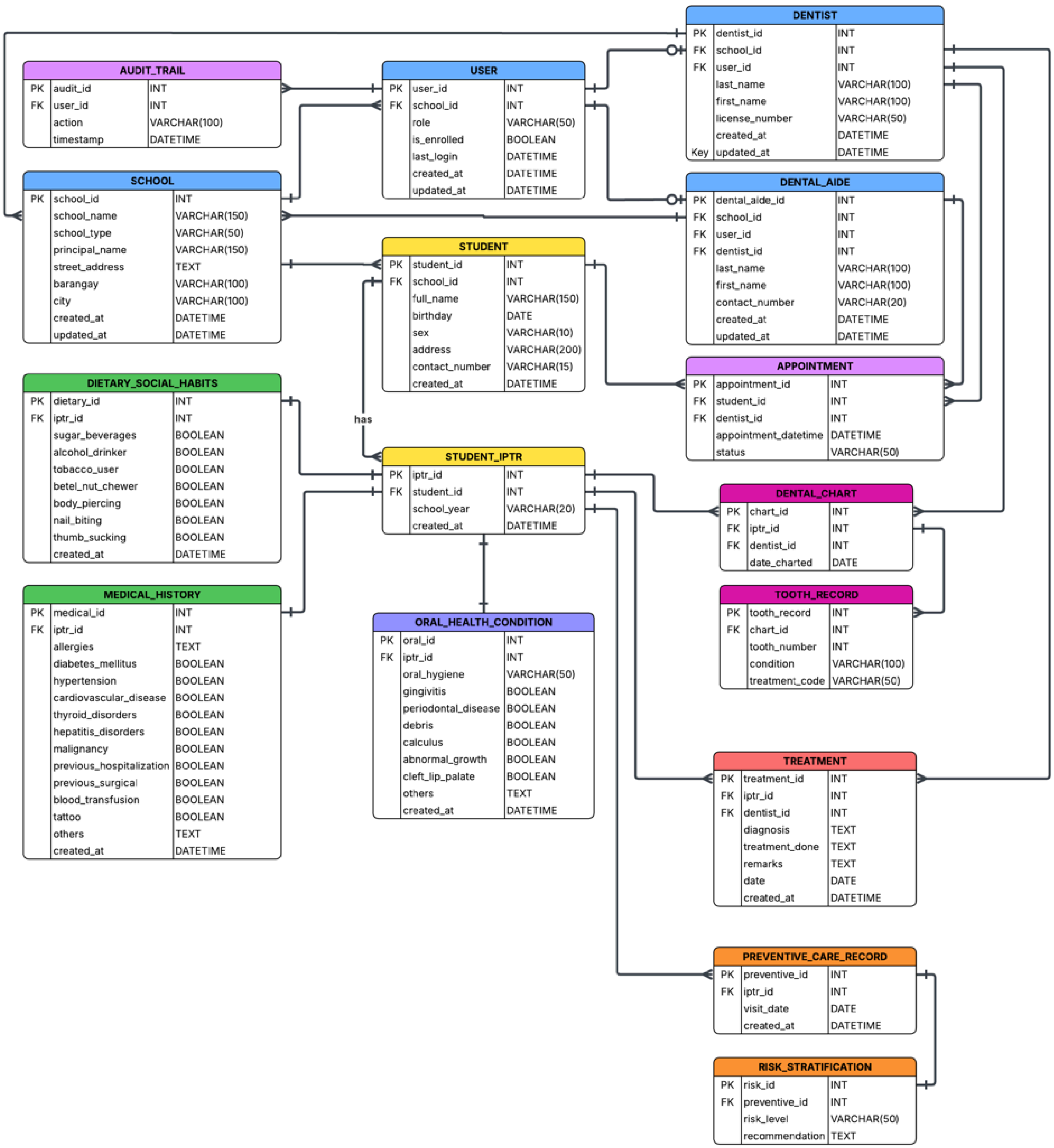

Figure 7. *Entity-Relationship Diagram*

Figure 7 presents the complete Entity-Relationship Diagram (ERD) of the proposed system, illustrating the logical structure of the centralized database and the relationships among its entities. The diagram follows a relational database design composed primarily of one-to-many relationships and selected one-to-one relationships, ensuring data integrity, scalability, and compliance with Third Normal Form (3NF).

The database is organized around three core entities: SCHOOL, USER, and STUDENT. The SCHOOL entity serves as the organizational reference, maintaining relationships with USER, STUDENT, DENTIST, and DENTAL_AIDE. Each school is associated with multiple users, and students, while a single dentist and dental aide may serve multiple schools, reflecting the real-world setup of shared dental personnel across campuses.

The USER entity manages system authentication and access control. It maintains a one-to-one relationship with both DENTIST and DENTAL_AIDE, ensuring that each dental staff member has a corresponding system account, while allowing other types of users (e.g., administrators) to exist independently.

The STUDENT entity represents the central subject of the system. Each student is linked to STUDENT_IPTR, which serves as the parent record for all dental health data per school year. This design centralizes and organizes health records while eliminating redundancy.

The STUDENT_IPTR entity maintains one-to-one and one-to-many relationships with several dependent entities, including MEDICAL_HISTORY, DIETARY_SOCIAL_HABITS, ORAL_HEALTH_CONDITION, DENTAL_CHART, TREATMENT, and PREVENTIVE_CARE_RECORD. These entities store categorized health information such as medical background, lifestyle habits, clinical findings, treatment history, and preventive care activities.

The DENTAL_CHART entity is further related to TOOTH_RECORD in a one-to-many relationship, allowing detailed recording of individual tooth conditions. Similarly, PREVENTIVE_CARE_RECORD is associated with RISK_STRATIFICATION to evaluate the student's oral health risk level and corresponding recommendations.

Additionally, the APPOINTMENT entity records scheduled consultations between students and dentists, while the AUDIT_TRAIL entity logs system activities performed by users for monitoring and security purposes.

Overall, the ERD demonstrates a well-structured and normalized database design that supports efficient data management, minimizes redundancy, and accurately represents the relationship within the school-based dental health monitoring system.

Population and Sample

The respondents of this study are the individuals who will directly use or receive outputs from the proposed system and who will evaluate its functionality suitability, performance efficiency, compatibility, usability, reliability, and security in accordance with the ISO/IEC 25010:2023 standard on software product quality [56]. Since the total number of identified respondents across all groups is small and manageable, the researchers adopted total population sampling [57], wherein all available and qualified individuals within each respondent group were included in the study. This approach eliminates the need for a sampling formula such as Slovin's formula, as the entire population is accessible and participation of all members is feasible.

Table 4. *Target Respondents*

Respondents	Total Number of Respondents	Target Respondents
Dentist	10	10
Dental Aides	10	10
School Clinic Staff	3	3
School Administrators	3	3
IT Evaluators	4	4
	Total	**30**

The dentists consist of 10 respondents, one of whom serves as the primary client of the system, being the assigned dental health personnel managing the three covered school dental clinics of Barangay Tanyag. The remaining nine dentists are personnel from other school dental clinics within Taguig City who participate in Bayanihan activities, where they are deployed to provide supplementary dental services in schools within Barangay Tanyag. Since these personnel will also be given access to the system during their deployment in the three covered schools, they are included as respondents to evaluate the system based on their actual interaction with it. This group will evaluate the system in terms of functional suitability, usability, and security based on their direct interaction with the patient record management, dental charting, appointment scheduling, and predictive analytics modules.

The dental aides similarly consist of 10 respondents, one of whom is the primary dental aide directly assigned to the three covered school dental clinics of Barangay Tanyag. The remaining nine dental aides are likewise personnel from other school dental clinics within Taguig City who are deployed during Bayanihan activities. As with the dentists, these dental aides will be given access to the system during their deployment and will evaluate the system in terms of functional suitability, usability, and security based on their direct interaction with the patient record management, appointment scheduling, and clinic coordination modules.

The school clinic staff consists of three respondents, referring to the nurses and other health personnel who work within the school clinic setting alongside the dentist and dental aide. In a clinic environment, these staff members also assist in attending to students' health complaints, supporting patient flow, and coordinating clinic activities. As personnel who are present in the clinic and interact with student health information in the course of their work, they will evaluate the system in terms of functional suitability and usability.

The school administrators consist of three respondents from Bagong Tanyag Integrated School, Bagong Tanyag Elementary School Annex A, and South Daang Hari Elementary School Main. As the administrative heads of the three covered schools, they are recipients of the system's automated reports and school-level dental health dashboards. This group will evaluate the system in terms of functional suitability and usability relative to their institutional oversight and reporting responsibilities.

The IT professionals consist of 10 respondents who will serve as independent technical validators of the system. This group will evaluate the system across all six quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, and security, based on their technical expertise, providing an objective and standards-based assessment of the system's overall software quality independent of their role as end users.

Sources of Data

The study will utilize multiple sources of data collection to ensure that both system requirements and evaluation results are accurate, relevant, and reliable. These sources will support the qualitative and quantitative components of the research.

Primary data will be collected directly from intended system users, particularly school dental personnel such as dentists, dental aides, and clinic staff. Interviews will be conducted to gather information about current workflows, challenges in managing dental records, and specific needs for improving patient care and record management. Observation will also be carried out within the dental clinic setting to examine how patient information is recorded, updated, and retrieved during daily operations. This will provide actual context on existing processes, including limitations of manual or paper-based systems.

Another source of data is the DOH’s ITPR. It is an official health document issued by the City Health Office of Taguig City that serves as a record of a patient's dental health history. It captures personal information such as the patient's name, birthday, age, sex, address, and PhilHealth details. The form also documents the patient's medical history, including known allergies, systemic conditions, and dietary and social habits that may influence oral health. It records oral health conditions identified during examination, covering conditions such as dental caries, gingivitis, periodontal disease, debris, calculus, abnormal growth, and others. The form includes a dental charting section that tracks the condition and treatment of individual teeth. Each clinical visit is also logged with details such as date, chief complaint, diagnosis, treatment done, and the attending dentist. This document will serve as a data source for the development of the system.

The Section-Level Oral Health Status Tally Form currently used will also be referenced in the study. This form documents the prevalence of dental conditions such as dental caries, gingivitis, debris, calculus, and DMF/dmf indices, broken down by sex per grade level or section.

The Target Client List for Oral Health Care and Services will also serve as a reference, as it contains individual patient entries with details such as date of consultation, name, address, date of birth, sex, age group, dental conditions observed, and treatments rendered.

The Oral Health Program Reporting Form provides consolidated data on oral health status indicators and services rendered across different age groups. This form will inform the design of the system's automated report generation, ensuring that the system's outputs are aligned with existing reporting requirements and formats.

The Parental Consent Form for School Dental Health Services used in the school dental clinics will also be referenced in the study as it reflects the scope of dental services offered to students, including oral examination, fluoride varnish application, oral prophylaxis, tooth restoration, and tooth extraction. This document provides context on the clinical workflow and the types of patient interactions that the system will need to support.

For the quantitative component, data will be collected through structured evaluation questionnaires administered to system users and IT evaluators. The questionnaire will be based on the ISO/IEC 25010:2023. Respondents will provide numerical ratings, which will be used for statistical analysis such as mean score computation.

Secondary data sources will include related literature, previous studies, and official documents relevant to dental health record management and predictive analytics. These materials will support the validation of system requirements and provide theoretical and technical foundations for the study.

By combining these sources of data collection, the study will ensure a comprehensive understanding of user needs, system performance, and predictive model effectiveness.

Research Locale

The study will be conducted in selected dental schools in Barangay Tanyag, Taguig City, Metro Manila, Philippines. Barangay Tanyag was selected as the research locale because it provides a practical and relevant setting for pilot testing a barangay-level dental health record management system. The barangay is well-suited for this study, as its school dental clinics have approximately 8,000 student records. This volume of data is sufficient to support the development, testing, and evaluation of the proposed system. Furthermore, the current reliance on manual and paper-based processes in these clinics highlights the urgent need for digitization, making it an ideal environment to assess the system's functionality.

Bagong Tanyag Integrated School, located in Purok 1, serves as the primary clinic site and the main station of the assigned school dentist. It is the only school among the three that offers both elementary and junior high school levels, making it the largest in terms of student population and the most operationally complex in terms of dental service delivery.

Bagong Tanyag Elementary School Annex A, located in Purok 13, is a public elementary school serving students from Kindergarten to Grade 6. It is one of the satellite schools visited by the rotating dentist and currently maintains its dental records entirely through manual, paper-based processes.

South Daang Hari Elementary School Main, located in Purok 11, is also a public elementary school that serves students from Kindergarten to Grade 6. Similar to Bagong Tanyag Elementary School Annex A, it relies on manual record-keeping and faces the same challenges of fragmented and difficult-to-access student dental data.

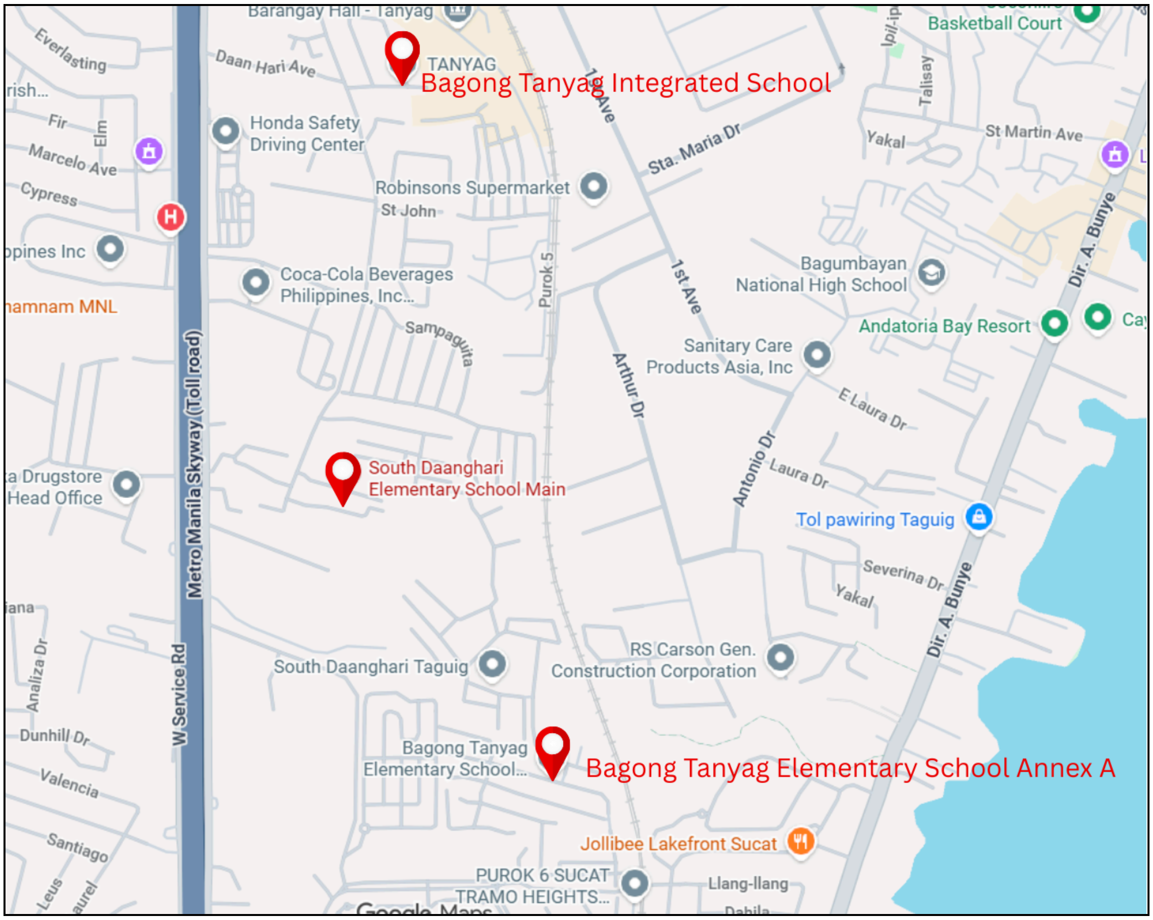

Figure 7. *Map of Barangay Tanyag, Taguig City showing the locations of the three covered school dental clinics.*

Method or Procedure

The study will follow a step-by-step process, including proposal approval, literature review, system design, development, testing, evaluation, and documentation.

Proposal Approval	The study begins with the preparation and submission of a formal research proposal to the research adviser and the panel of examiners. The proposal includes the research problem, objectives, scope and limitations, significance of the study, and an initial

	conceptual framework. Upon approval, the researchers proceed to the planning and development phases.
Literature Review and Requirements Gathering	The researchers conduct a comprehensive review of related literature and studies on dental health record management, digital dental charting systems, predictive analytics in oral health, and existing school-based dental programs. To gather system requirements, the researchers conduct interviews and consultations with the client from the Barangay Tanyag school dental clinics, including dentists, dental aides, clinic staff, and school administrators. These aim to understand current manual processes and identify pain points for the proposed system.
System Design and Sprint Planning	Following the Agile methodology, the researchers will break down the development work into manageable sprints. Each sprint will focus on specific system modules: user authentication and role management, student dental record management, digital dental charting with standard dental notation, appointment scheduling and monitoring, routine preventive care tracking, predictive analytics integration, and dashboard and report generation. The proposed system will be documented using system architecture, context diagrams, top-level data flow diagrams, entity-relationship diagrams, and use case diagrams to ensure a clear and structured design.
Development Tools	The system will be developed iteratively using the MERN stack, which consists of MongoDB for database management, Express.js as the backend framework, React for the frontend user interface, and Node.js as the runtime environment. The development will follow an Agile methodology, with each sprint delivering functional modules that will be tested and refined incrementally. For the predictive analytics module will be developed using Python-based machine learning libraries such as scikit-learn, pandas, and numpy, integrated with the MERN backend via API endpoints to classify students

	into High, Medium, or Low oral health risk based on historical dental records. The modular architecture of the MERN stack will allow for efficient development, seamless integration of components, and scalable deployment across the three covered schools in Barangay Tanyag.
Sprint Testing and Debugging	At the end of each sprint, the developed modules will undergo unit testing and integration testing to verify that individual components function correctly and that they work seamlessly with existing modules. Identified bugs and issues will be documented and resolved before proceeding to subsequent sprints. This iterative testing will ensure that the system maintains stability and reliability throughout the development process.
Pilot Testing and Reliability Assessment	Before full-scale evaluation, the researchers will conduct a pilot test with a small group of end-users, including selected dentists, dental aides, IT professionals, and school administrators. The pilot test will aim to identify usability issues, technical glitches, and areas for improvement. For system validation, the researchers will utilize a researcher-developed questionnaire based on ISO 25010:2023 quality characteristics. Since not all items in the validation instrument may be directly applicable to the study context, a reliability test will be conducted to assess the internal consistency of the instrument using Cronbach's Alpha and similar tools.
User Acceptance Evaluation	Following the pilot test and necessary refinements, the system will undergo formal evaluation by the target respondents, including dentists, dental aides, clinic staff, school administrators.. The evaluation will use the ISO 25010:2023 based questionnaire to assess the system across six quality characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, and Security. Responses will be collected using a five-point Likert scale, and weighted means will be computed to

	determine the overall performance rating of the system.
Data Analysis	The collected evaluation data will be analyzed using descriptive statistical tools, including weighted mean, frequency count, and percentage. The results will be interpreted based on established scale ranges to determine the level of user satisfaction and system effectiveness. Findings from the evaluation will inform the final refinements to the system.
System Finalization	Based on the feedback and evaluation results, the researchers will implement final adjustments to the system. These refinements may include user interface enhancements, performance optimizations, and additional features identified as critical during the evaluation phase. The final version of the system will then be prepared for deployment.
Documentation and Recommendation	Upon completion of system development and evaluation, the researchers will document the entire project, including the development process, evaluation results, and findings. The study will conclude with recommendations for future enhancements.

Instruments and Data Collection

The study utilized a combination of qualitative and quantitative data-gathering instruments to support the development and evaluation of the Dental Health Record Management System with Predictive Analytics.

Interview

A semi-structured interview was conducted with the client, particularly the school dentist and dental aide, to gather qualitative data regarding the current workflow, dental charting practices, and challenges in managing dental records. The interview included open-ended questions that allowed the respondents to describe existing processes, system needs, and operational concerns.

In addition, an interview guide, administered in a questionnaire format, was used to determine whether specific attributes from the dental record form are contributing factors to various oral health conditions.This process ensured that the variables used in the predictive analytics component are aligned with actual clinical practices.

Survey Questionnaire

A researcher-developed questionnaire will be utilized to evaluate the prototype of the system. The questionnaire will focus on assessing the system's functional suitability, performance efficiency, compatibility, usability, reliability, security based on selected criteria from ISO/IEC 25010:2023.

System Demonstration

A prototype of the proposed Dental Health Record Management System with Predictive Analytics will be presented to the respondents during the data collection process. The demonstration allowed users to explore the system's features, including patient records, dental charting, and scheduling. This ensured that the responses gathered were based on actual interaction with the prototype rather than assumptions.

Data Collection Procedure

The data collection was conducted through direct interaction with the respondents. The researchers first conducted the interview to gather qualitative insights and validate system requirements. The prototype was then demonstrated to the respondents, followed by the administration of the ISO/IEC 25010:2023 questionnaires for system evaluation. The collected data will then be analyzed to assess the system's effectiveness and identify areas for improvement.

Statistical Treatment

The researchers will employ descriptive statistical analysis tools, specifically weighted mean, frequency count, and percentage, to analyze the responses gathered from the evaluation of the Dental Health Record Management System.

Weighted Mean

The researchers used the weighted mean to determine the average perception of respondents for each criterion evaluated using the Likert scale [58]. This method assigns importance to each response by multiplying the frequency of each rating by its corresponding weight. The formula is as follows:

$$X_{\omega} = \frac{\sum(X_i \cdot W_i)}{\sum W_i}$$

Figure 8. *Weighted Arithmetic Formula*

Where:

$\bar{x}_{\omega}$ = weighted mean

x_i = the response value (e.g., 5, 4, 3, 2, 1)

ω_i = the frequency or number of respondents who selected that response

Frequency Count and Percentage

The researchers used frequency count and percentage to determine the distribution of responses and the proportion of respondents who selected each rating on the evaluation scale. A percentage is a number or ratio expressed as a fraction of 100, denoted using the percent sign (%) [58].

To get the rate, the researchers used the percentage formula:

Percentage =($Frequency\ of\ Response/Total\ Number\ of\ Respondents$) x 100

Likert Scale

The researchers will use the Likert scale to assess respondents' perceptions and satisfaction with the system's functionality, usability, reliability, security, performance efficiency, and compatibility [59]. Participants will rate their level of agreement with various statements regarding the system's features and performance using a five-point scale, ranging from Strongly Disagree to Strongly Agree, which allows the researchers to systematically quantify user feedback and determine areas for improvement based on the weighted mean of responses [60].

Table 5. *Five-point Likert Scale*

Response	Value
Strongly Agree	5
Agree	4
Neutral	3
Disagree	2
Strongly Disagree	1

Scale Interpretation

The researchers will use the scale interpretation table to determine the meaning of the weighted mean scores obtained from the Likert scale responses [61]. The interval of 0.80 was derived by dividing the total scale range of 4 by the number of response categories, which is 5. This allows the researchers to systematically interpret the level of agreement or disagreement of respondents with respect to each evaluated criterion of the proposed system. The mean scores will be interpreted based on the range corresponding to each verbal interpretation as follows:

Table 6. *Scale Interpretation*

Mean Range	Verbal Interpretation
4.21 - 5.00	Strongly Agree
3.41 - 4.20	Agree
2.61 - 3.40	Neutral
1.81 - 2.60	Disagree
1.00 - 1.80	Strongly Disagree

The researchers will use Microsoft Excel, Google Sheets, Python, or R as a tool for data analysis and visualization due to its ability to organize, compute, and present numerical data in a clear and structured format. This allows for an accurate and comprehensible presentation of the evaluation results gathered from the respondents. To achieve the desired results from the gathered data, the researchers will employ various statistical formulas to tabulate and calculate the numerical data.